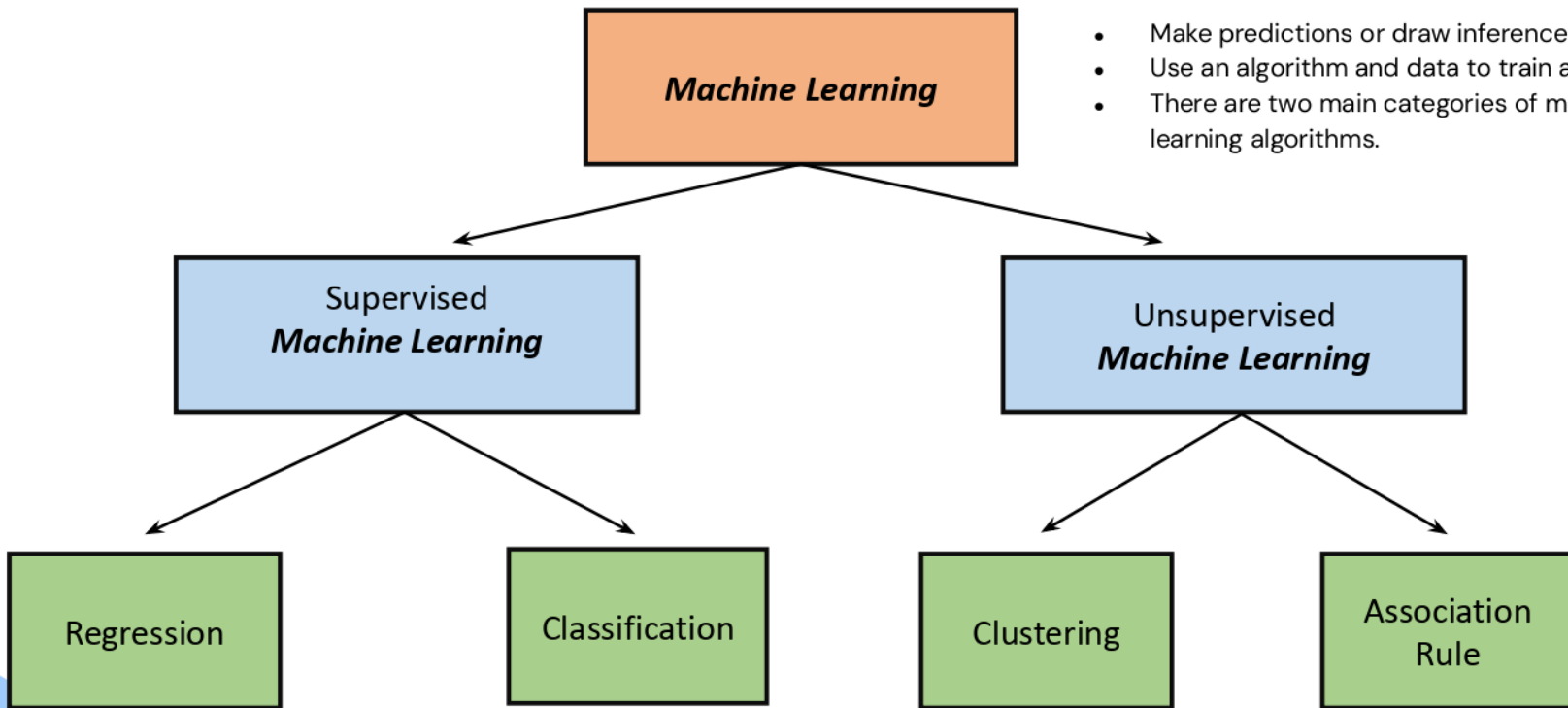

Linear Regression

Édipo Cremon



Machine Learning Models

- Make predictions or draw inferences.
- Use an algorithm and data to train a model.
- There are two main categories of machine learning algorithms.



Machine Learning Models

Machine Learning

- Make predictions or draw inferences.
- Use an algorithm and data to train a model.
- There are two main categories of machine learning algorithms.

Supervised
Machine Learning

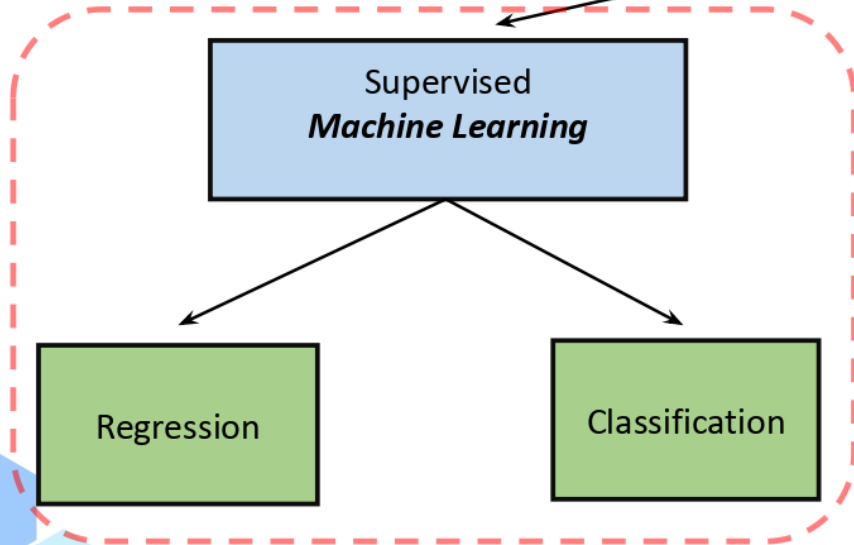
Regression

Classification

Unsupervised
Machine Learning

Clustering

Association
Rule



Supervised Machine Learning

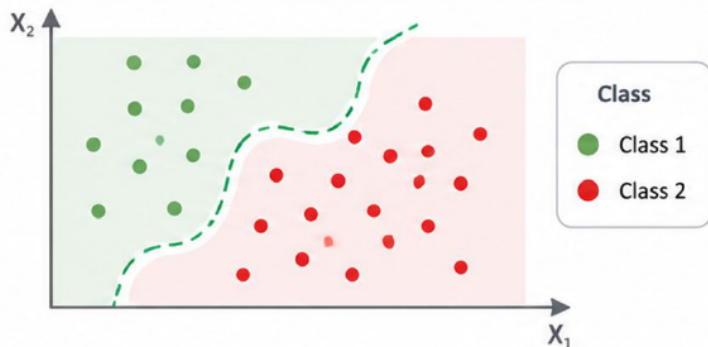
What is the difference?



Classification



Predicts a class or category
(discrete output)



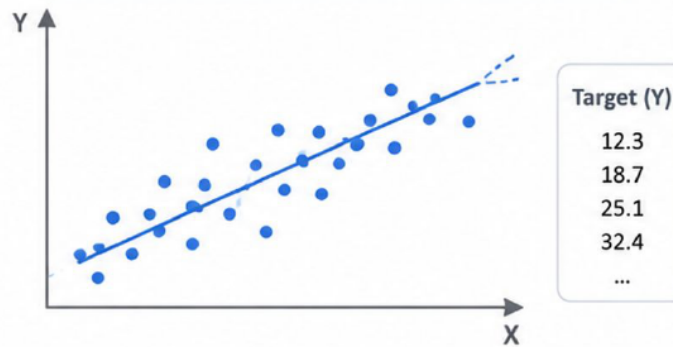
Examples: Spam detection, disease diagnosis, image recognition, sentiment analysis



Regression



Predicts a continuous numerical value
(continuous output)



Examples: House price prediction, sales forecasting, temperature prediction, demand estimation



Key takeaway: Classification answers “which class?” | Regression answers “how much?”

Variable Relationships

Understanding how variables are related help us build models and make predictions



Each row represents the same student

Student	Hours Studied (per week)	Attendance (%)	Sleep (hours per night)	Exam Score (%)
1	1.1	65	5.7	55
2	2.5	72	6.1	63
3	3.2	80	6.8	72
4	4.0	85	7.0	78
5	4.6	90	6.5	85
6	5.1	92	7.3	88
7	6.3	95	6.4	92
8	6.8	96	7.6	94
9	7.5	97	6.2	96
10	8.0	98	7.1	97

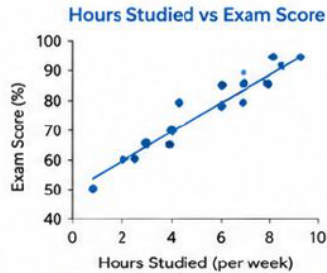


Goal:

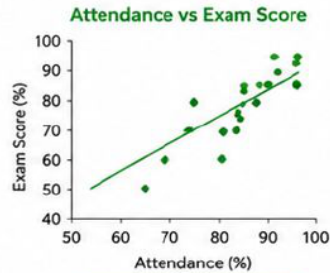
Understand the relationship among variables to develop models that can predict the behavior of one variable from the values of one or more other variables.

Variables

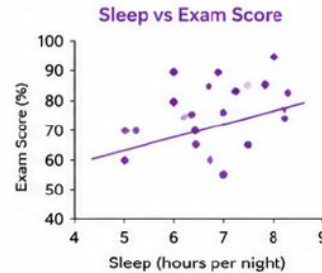
- Hour studied (per week)
- Attendance (%)
- Sleep (hours per night)
- Exam Score (%) (target)



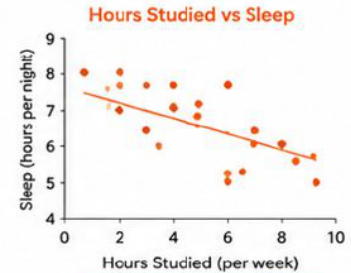
Strong positive relationship



Positive relationship



Weak relationship



Negative relationship

Key takeaway: Different variables can be related in different ways. By quantifying these relation, we can build models to predict outcomes such as exam perf.

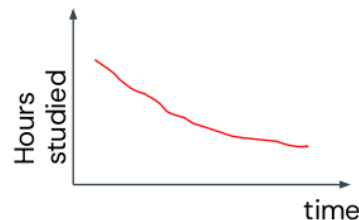
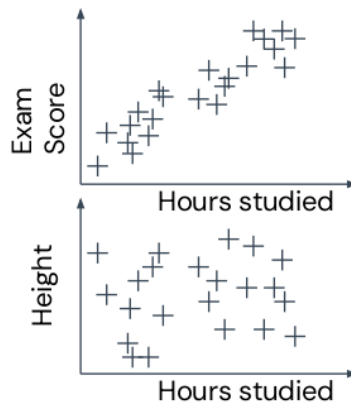
Variable Relationships

Exemple:

Consider the variable **Hours Studied**, which is positively related to **Exam Score**. Students who study more hours tend to achieve higher exam scores. On the other hand, **Hours Studied** has no relationship with a student's **Height**.

If we observe a decrease in the number of hours studied over time, what can we conclude about the student's exam performance and height?

- **Exam Score:** A decrease in exam performance is expected.
- **Height:** No conclusion can be drawn.



Correlation



Indicates the **linear relationship** between two variables.



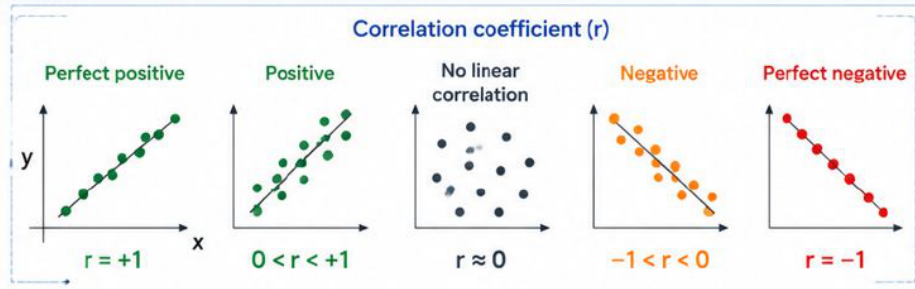
It's a measure of the relationship between two attributes, although correlation does **not imply causation**.



- Two variables may be highly correlated without a cause-and-effect relationship between them.



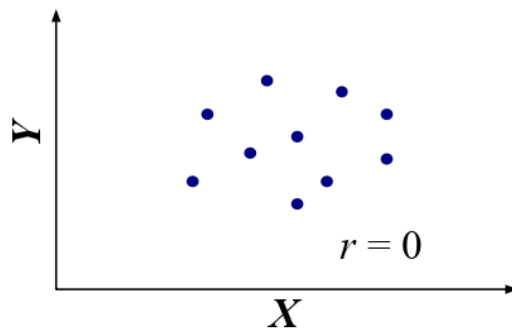
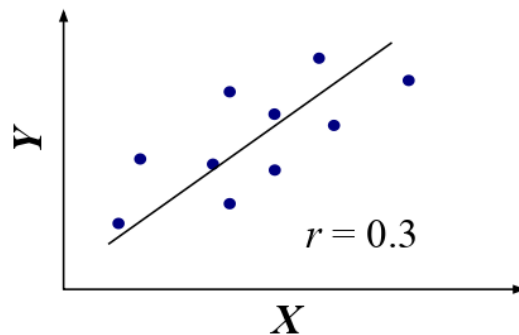
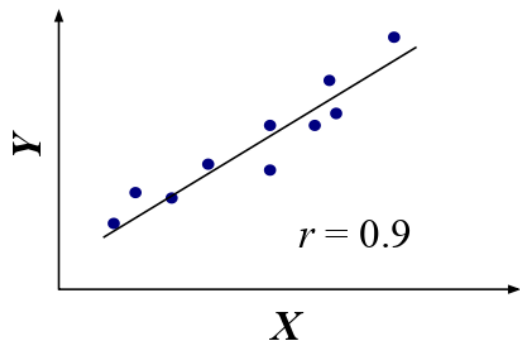
Two or more variables **may be related**, making it necessary to investigate the nature of this relationship.



- A correlation coefficient close to +1 or -1 indicates a **linear relationship** between two attributes, making it possible to **fit a model** that represents this relationship.
- This is the objective of **regression analysis**.

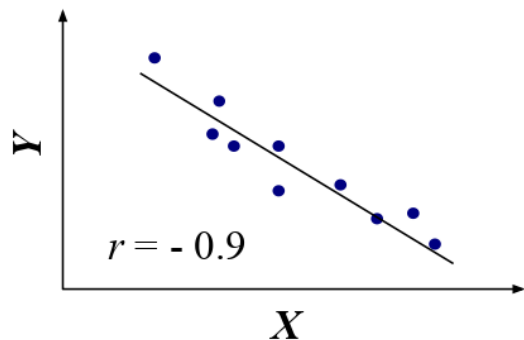
How can the degree of relationship (or association) between pairs of variables be characterized?

Correlation Coefficient



Pearson Correlation Coefficient

Measures the strength of the **linear** relationship between X and Y.



$$r = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

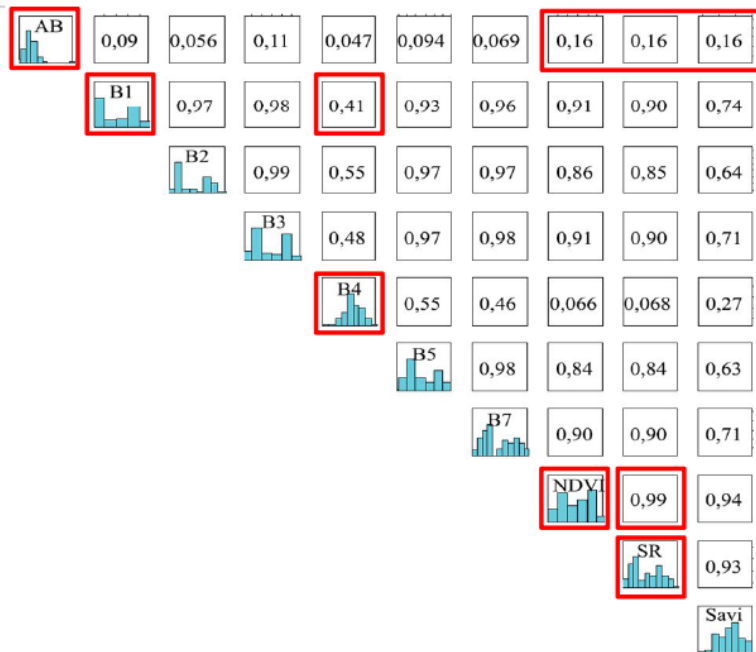
$$r = \frac{\sum_{i=1}^n (x_i - \bar{X})(y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{X})^2 \sum_{i=1}^n (y_i - \bar{Y})^2}}$$

$$-1 \leq r \leq 1 \quad (\text{Dimensionless})$$

$$r = \frac{n \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i \sum_{i=1}^n y_i}{\sqrt{\left[n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2 \right] \left[n \sum_{i=1}^n y_i^2 - \left(\sum_{i=1}^n y_i \right)^2 \right]}}$$

Correlation Coefficient

Correlations (absolute values) between dendrometric characteristics of the Brazilian Caatinga and Landsat 5 TM data (Almeida et al., 2014*)



AB - basal area

B1 a B7 – Landsat TM spectral bands

$NDVI = (B4 - B3) / (B4 + B3)$

$SR = B4 / B3$

$Savi = 0,5(B4 - B3) / (B4 + B3 + 0,5)$

The **correlation coefficient** measures the strength of the linear relationship between **X** and **Y**.

→ it is essential to examine the **scatter plot** to verify whether the relationship is truly linear and whether the data points are appropriately distributed.

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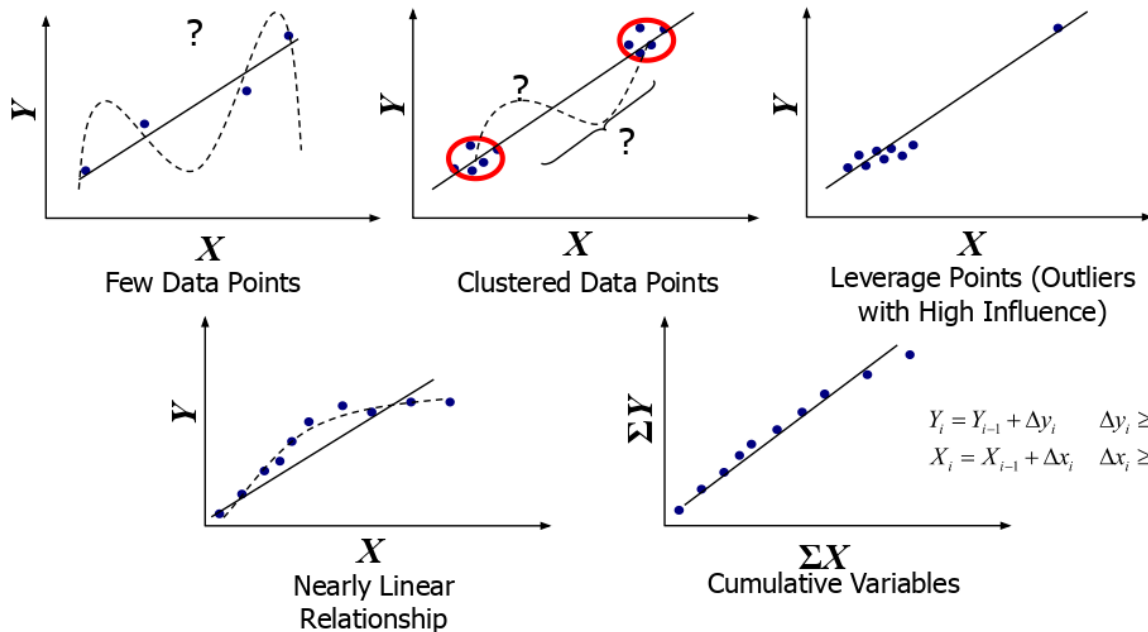
→ it is essential to examine the **scatter plot** to verify whether the relationship is truly linear and whether the data points are appropriately distributed.

*Source: <http://www.scielo.br/pdf/pab/v49n4/0100-204X-pab-49-04-0306.pdf>

Correlation Coefficient

Common Misinterpretations of the Correlation Coefficient

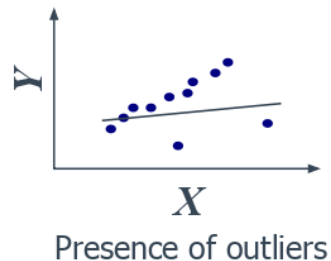
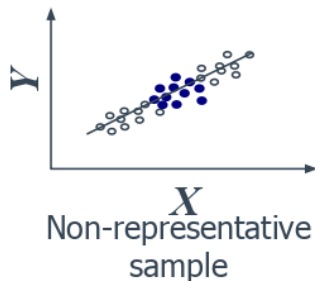
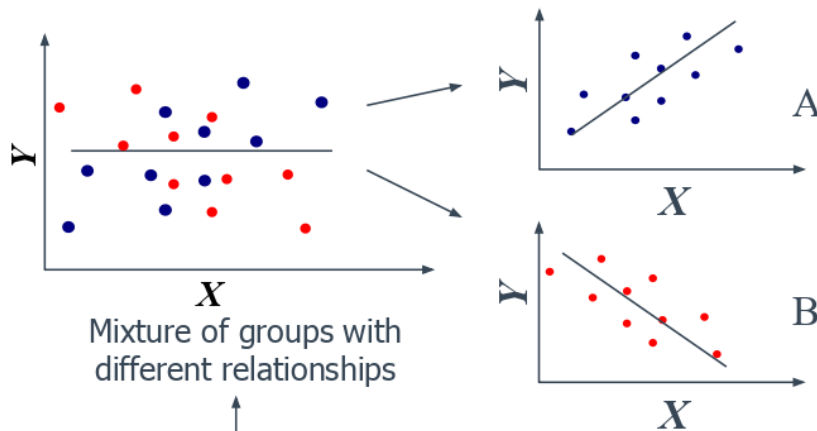
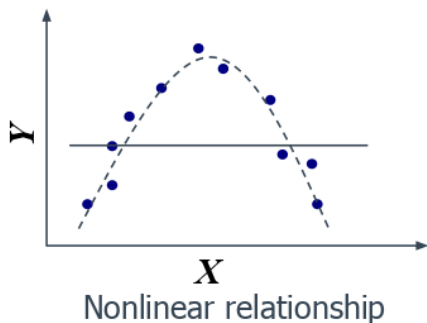
A high correlation coefficient (in absolute value) does not always indicate that the estimated regression equation provides a good fit to the data.



Correlation Coefficient

Common Misinterpretations of the Correlation Coefficient

A high correlation coefficient (in absolute value) does not always indicate that the estimated regression equation provides a good fit to the data.





Linear Regression Analysis

Regression Analysis

What is regression analysis used for?

- Identify the variables that are most strongly related to the **dependent variable (Y)**.
- Determine the function that describes how one or more variables are related to the **dependent variable (Y)** and estimate the parameters that define this function (the fitted equation).
- Use the fitted equation to predict values of the **dependent variable (Y)**.

Simple Linear Regression

Simple Linear Regression Model

Assumptions:

$$E(\varepsilon_i) = 0$$

$$\text{Var}(\varepsilon_i) = \sigma^2$$

$$\text{Cov}(\varepsilon_k, \varepsilon_j) = 0 \quad \forall k \neq j$$

independent errors

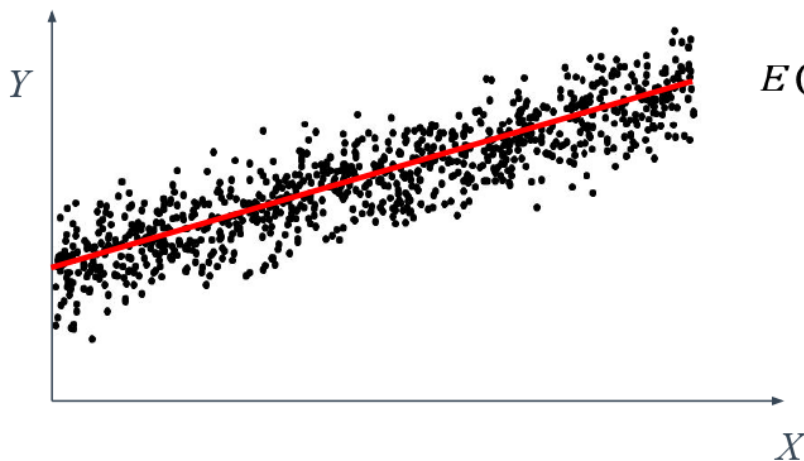
$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

Random component
(error or residual)

Independent variable*
(predictor variable)

Dependent variable
(response variable)

β_0 and β_1 are fixed parameters.

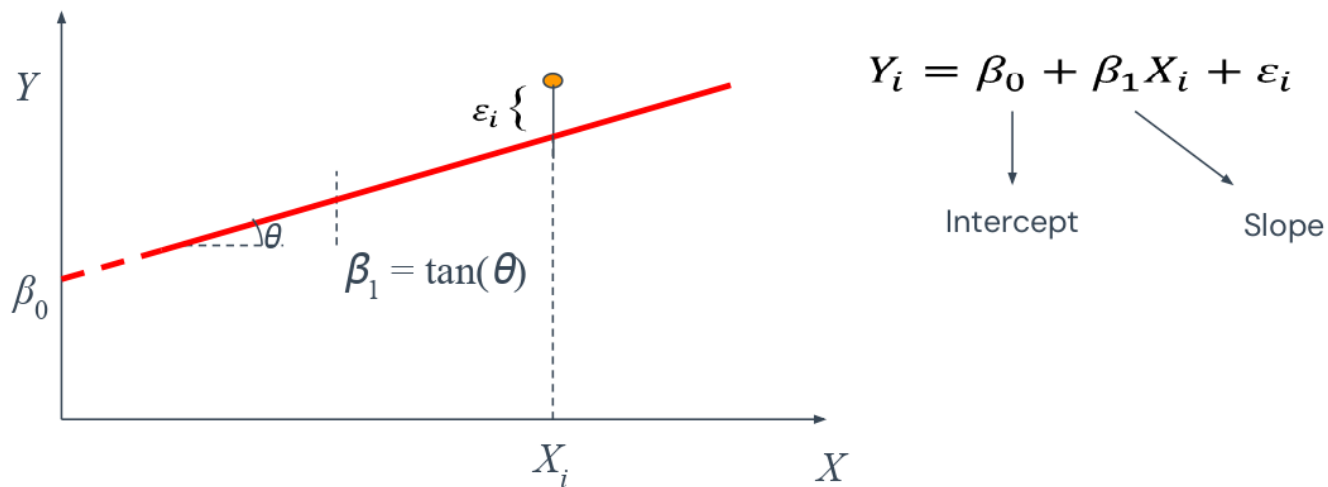


$$E(Y_i) = \beta_0 + \beta_1 X_i$$

The regression line represents the **mean value of the dependent variable (Y)** for each level of the **independent variable (X)**.

* In classical regression, the **independent variable** is not treated as a random variable; that is, its values are assumed to be fixed and known.

Simple Linear Regression Model



β_0 represents the value of Y_i when $X_i = 0$.

β_1 is the slope of the regression line and represents the change in Y_i associated with a one-unit increase in X_i .

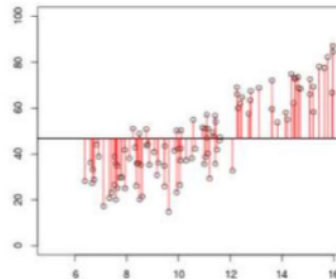
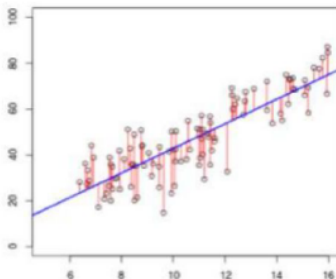


Characteristics of a Regression Model

Characteristics of a Regression Model

A regression model essentially describes:

- The tendency of Y to vary systematically with the predictor X .
- The dispersion of data points around the curve that represents the statistical relationship.

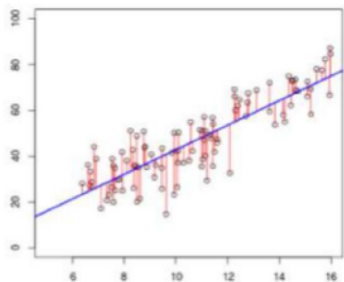


These characteristics are expressed through the following assumptions:

- There is a probability distribution of Y for each level (value) of X .
- The mean of these distributions varies systematically with X .

Characteristics of a Regression Model

- We assume that, for a given value of X , the variable Y follows a probability distribution whose expected value is given by $E(Y_i)$.
- The expected value of the errors is zero ($E(\varepsilon) = 0$):



$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

↓ ↘
Intercept Slope

The regression line represents the **mean value of the dependent variable (Y)** for every level of the **independent variable (X)**.

- The error variance is constant, meaning that the errors are homoscedastic.

The errors are:

- Not autocorrelated:
- Mutually independent
- Normally distributed:

$$(\text{Cov}(\varepsilon_i, \varepsilon_j) = 0), i \neq j$$

$$\varepsilon_i \sim N(0, \sigma^2)$$

Goodness of Fit: Coefficient of Determination (R^2)

- **Coefficient of Determination (R^2):** Measures the proportion of the variation in the **dependent variable** that is explained by the **independent variable** in the regression model.

$$R^2 = \frac{SS_{Reg}}{SST} = \frac{\sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2}$$

SST (Total Sum of Squares): Total variation in the dependent variable.

SSR (Residual Sum of Squares): Unexplained variation (sum of squared residuals).

SSReg (Regression Sum of Squares): Explained variation (sum of squares due to regression).

Note: R^2 ranges from **0** to **1**. Values close to **1** indicate that the model provides a good fit and adequately describes the phenomenon.

Goodness of Fit: Coefficient of Determination (R^2)

How can we interpret the value of R^2 ?



R^2 represents the proportion (or percentage) of the total variance in Y that is explained by the regression model.

Note: In regression analysis, R^2 should always be interpreted alongside other performance metrics to obtain a more comprehensive evaluation of the model's performance.



Correlation Coefficient (R)

$\neq$

Coefficient of Determination (R^2)

Linearizable Models

Standard Model: $Y_i = \beta_0 + \beta_1 X_i + \xi_i$

exponential

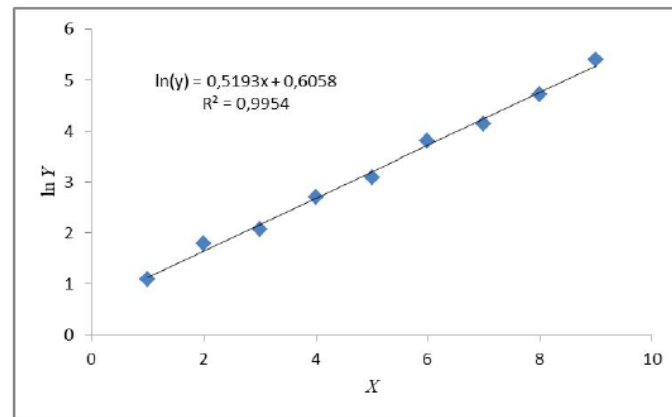
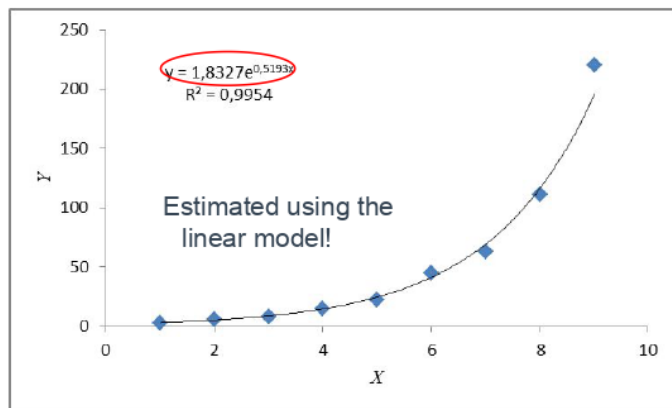
$$Y_i = \beta_0 e^{\beta_1 X_i} \xi_i$$

$$\ln Y_i = \ln \beta_0 + \beta_1 X_i + \ln \xi_i$$

$$Y'_i = \beta'_0 + \beta_1 X_i + \xi'_i$$

$$\xi'_i \sim N(0, \sigma^2)$$

exponential



Linearizable Models

Standard Model: $Y_i = \beta_0 + \beta_1 X_i + \xi_i$

potential

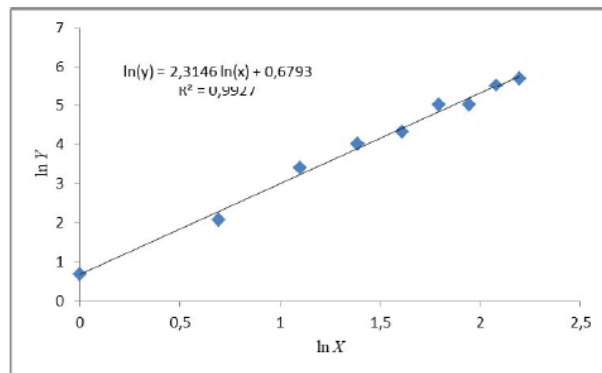
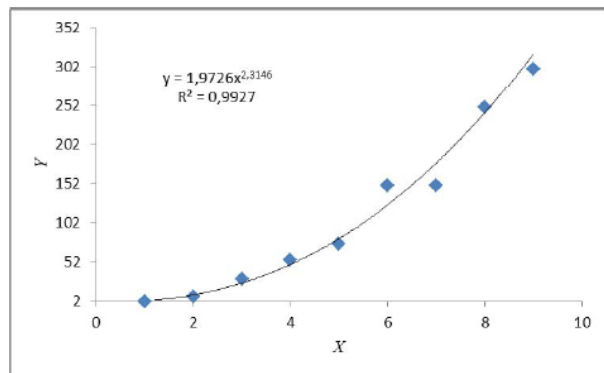
$$Y_i = \beta_0 X_i^{\beta_1} \xi_i$$

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$$Y'_i = \beta'_0 + \beta'_1 X'_i + \xi'_i$$

$$\xi'_i \sim N(0, \sigma^2)$$

potential



Applying a series of transformations (e.g., logarithmic, power, reciprocal) to **X**, **Y**, or both allows the **simple linear regression model** to represent more complex relationships. $\Rightarrow$ **Generalized Linear Models (GLMs)**

Simple Linear Regression Model

$$Y = \beta_0 + \beta_1/\beta_2 X$$

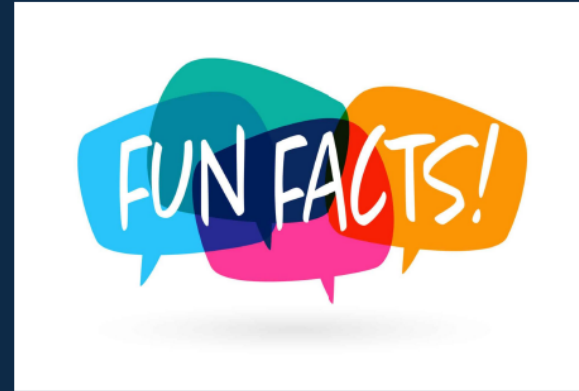
is not a linear combination

$$Y = \beta_0 + e^{\beta_1} X$$

is not a linear combination

Linearity refers to the parameters, not the variables.

Correlation is a **statistical measure** that quantifies the strength and direction of the relationship between two (or more) variables. It is expressed by values ranging from **-1 to 1**. Positive values closer to **1** indicate a positive association between the variables, meaning that as one variable increases, the other also tends to increase.

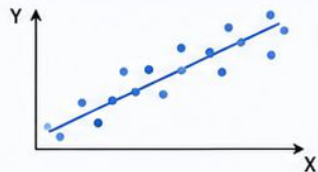


In contrast, the term **relationship** (or **association**) can be used to describe **any type of association** between two or more variables. It is a **broader concept** and is not limited to statistics.

Be careful not to use the term "correlation" incorrectly.

Assumptions of Linear Regression

1 Linearity



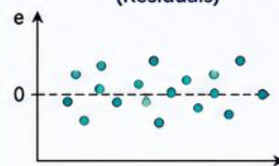
The relationship between the **dependent variable** and the **independent variable(s)** must be linear.

2 Accurate Measurement



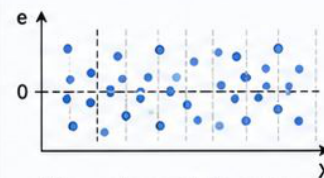
The variables must be **measured accurately**; no systematic measurement error is assumed.

3 Zero Mean of Errors (Residuals)



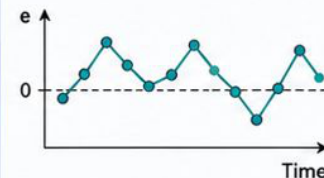
The expected value of the **error term (residuals)** is zero.

4 Homoscedasticity



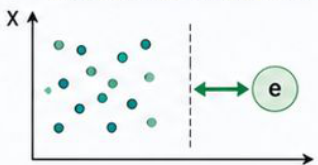
The variance of the error term is **constant across** all values of the independent variable(s).

5 No Autocorrelation



The error terms are **independent** of one another.

6 Exogeneity (No Correlation with Errors)



The independent variable(s) must **not be correlated** with the error term.

7 No Omitted Relevant Variables



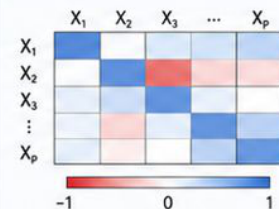
Include relevant variables



Omit important variables

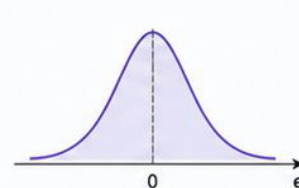
No theoretically **relevant** variable for explaining Y should be **omitted** from the model, and no **irrelevant** variable should be included.

8 No Multicollinearity



The independent variables must not exhibit **high correlation** with one another.

9 Normality of Errors



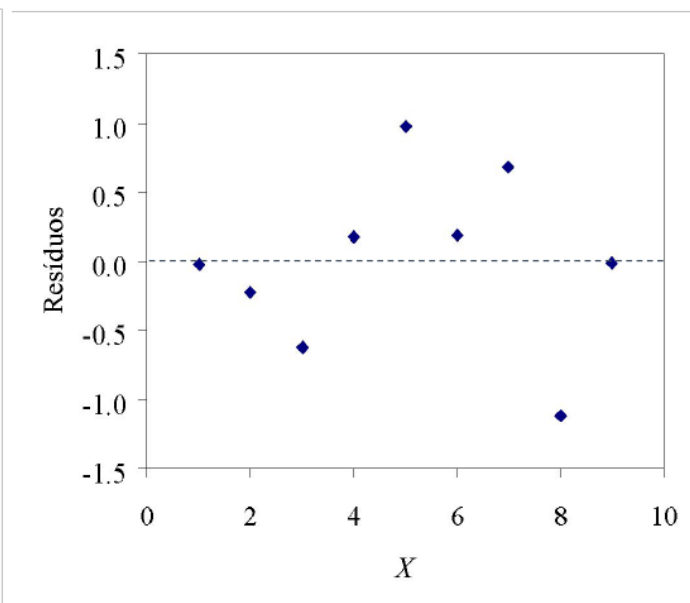
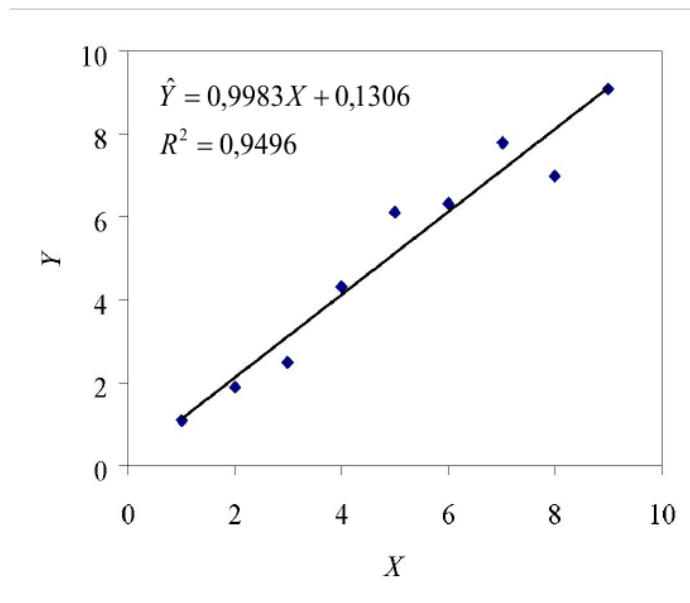
The error term is assumed to follow a **normal distribution**.

10 Adequate Sample Size



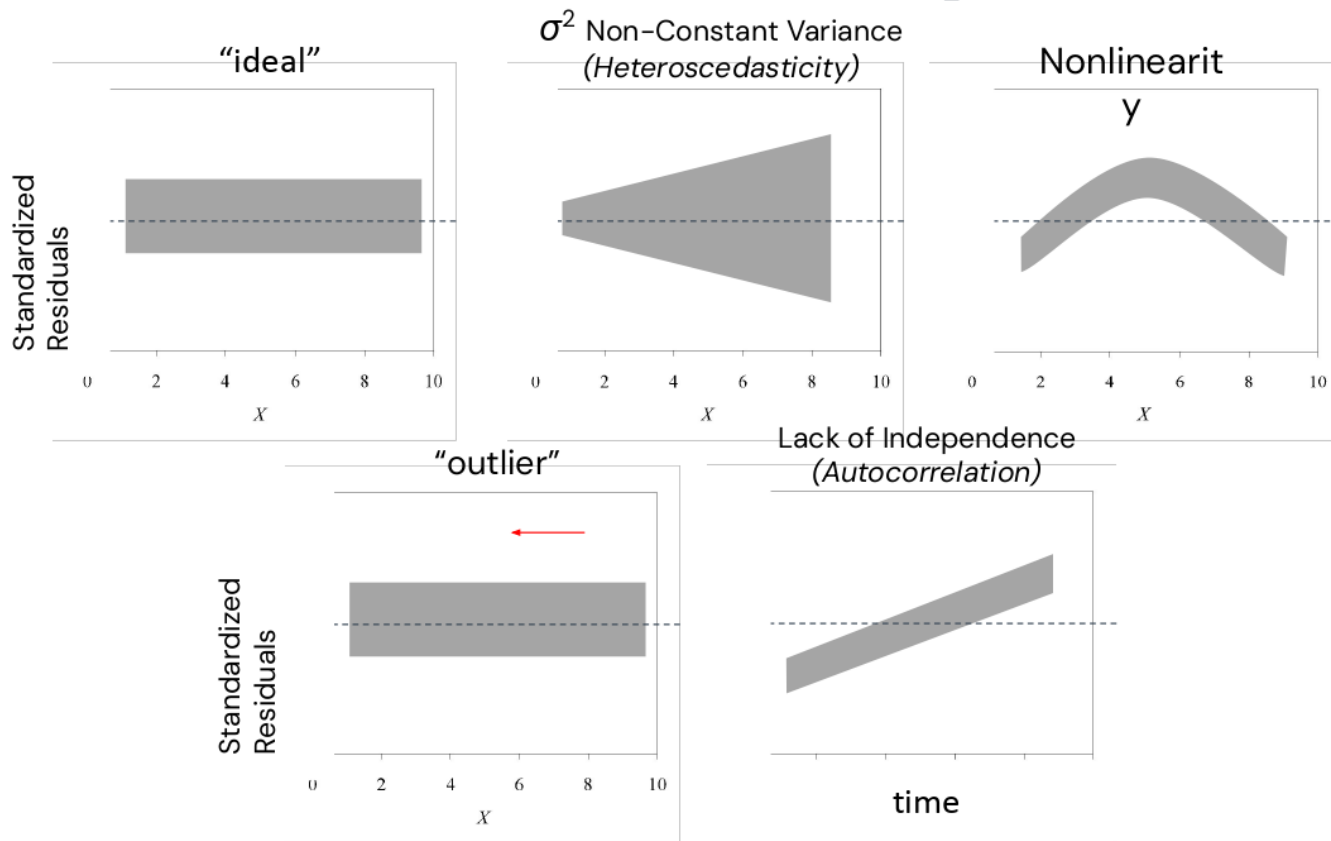
There must be an adequate ratio between the number of **observations (n)** and the number of **estimated parameters (p)**.

Residual Analysis



$$\text{Residual} = e_i = Y_i - \hat{Y}_i$$

Residual Analysis



Regression Model Goodness of Fit

Regression performance metrics are based on the difference between the **observed** and **predicted** values:

$$e = y - \hat{y}$$

- **Hypothesis Testing** for the model parameters
- **ANOVA (Analysis of Variance)**
- **MSE (Mean Squared Error)**
- **RMSE (Root Mean Squared Error)**
- **MAE (Mean Absolute Error)**
- **MAPE (Mean Absolute Percentage Error)**
- **RMSLE (Root Mean Squared Logarithmic Error)**

Regression Model Goodness of Fit

Mean Squared Error (MSE)

The **Mean Squared Error (MSE)** is the average of the squared prediction errors. In practice, it measures the **average squared prediction error**:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$$

- The **higher** the MSE, the **worse** the model.
- Its **minimum value is 0**, and it has **no upper bound**.

Regression Model Goodness of Fit

Root Mean Squared Error (RMSE)

The **RMSE** measures the predictive performance of a regression model. It quantifies the average difference between the **predicted values** and the **observed values**:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2}$$

- It can be interpreted as the **average deviation** of the predictions from the target values.
- The **lower** the RMSE, the **better** the model's predictive performance.

Regression Model Goodness of Fit

Mean Absolute Error (MAE)

The **Mean Absolute Error (MAE)** is calculated by taking the **absolute error** for each prediction:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i|$$

- The **lower** the MAE, the **better** the model's performance.
- Its **minimum value is 0**, and it has **no upper bound**.

Regression Model Goodness of Fit

Mean Absolute Percentage Error (MAPE)

The **MAPE** is a percentage-based metric obtained by dividing the absolute difference between the **predicted value** ($\hat{y}$) and the **observed value** (y) by the observed value:

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

- A **low MAPE** indicates that the predictions are close to the observed values, meaning the model performs well.
- A **high MAPE** indicates that the predictions deviate substantially from the observed values, suggesting that the model needs improvement.

Regression Model Goodness of Fit

Root Mean Squared Logarithmic Error (RMSLE)

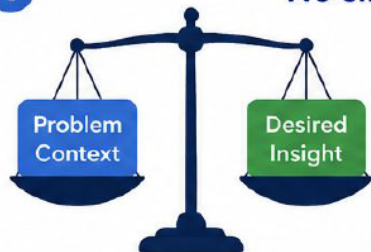
- The **RMSLE** is computed similarly to the **RMSE**.
- The logarithmic transformation reduces the impact of large differences between the **predicted** and **observed** values when both values are very large.

$$\text{RMSLE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\log(y_i + 1) - (\log \hat{y}_i + 1))}$$

Regression Model Goodness of Fit

1

No single "best" metric

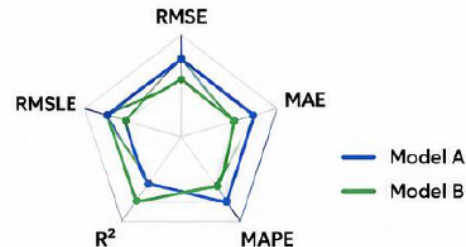


The most appropriate metric depends on the problem and on the type of insight you want to obtain.

2

Use multiple evaluation metrics

Multiple metrics reveal different strengths and weaknesses of the model.

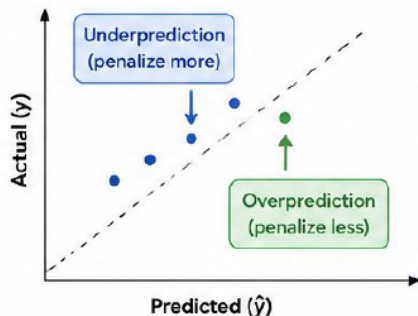


3

Define your own evaluation metric

You can define your own metric if it is better suited to your application.

Example: penalize **underpredictions** differently from **overpredictions**.



4

Follow the MAPE approach

To create a custom evaluation metric, simply follow the same approach used to compute the **MAPE**.



$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100\%$$



Key Takeaway: Choose metrics that align with your goal, use multiple metrics, or define a custom one when needed.

Multiple Linear Regression

General Model

$$Y_i = \beta_0 + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \dots + \beta_{p-1} X_{p-1,i} + \xi_i$$

$\beta_0, \beta_1, \beta_2, \dots, \beta_{p-1}$ are the model parameters (p parameters in total)

$X_{1,i}, X_{2,i}, \dots, X_{p-1,i}$ are fixed, known values

ξ_i are independent errors, with $\xi_i \sim N(0, \sigma^2)$

$i = 1, 2, \dots, n$

By setting $X_{0,i} = 1$, the model can be rewritten as:

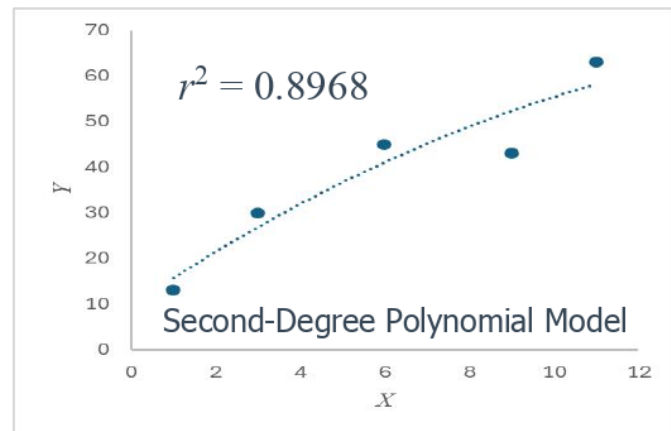
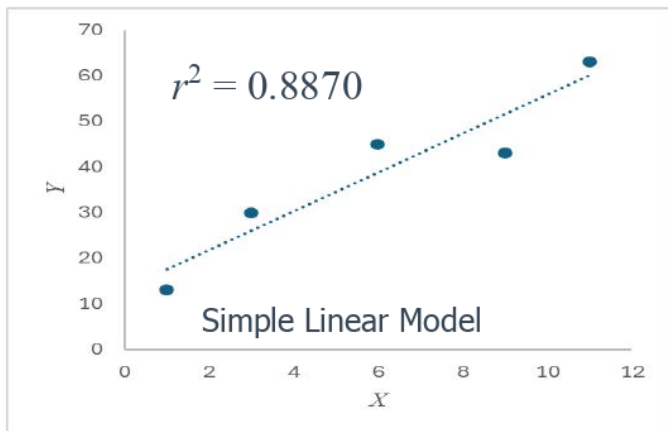
$$\begin{aligned} Y_i &= \beta_0 X_{0,i} + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \dots + \beta_{p-1} X_{p-1,i} + \xi_i \\ &= \sum_{k=0}^{p-1} \beta_k X_{k,i} + \xi_i \end{aligned}$$

Example: Tomorrow's rainfall (mm) can be predicted as a function of atmospheric pressure, air temperature, and today's rainfall.

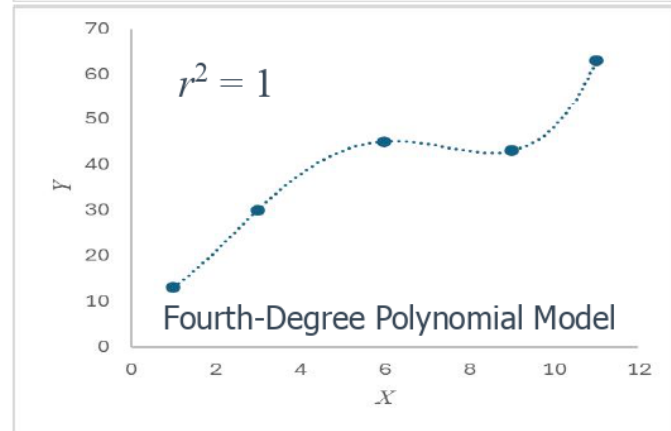
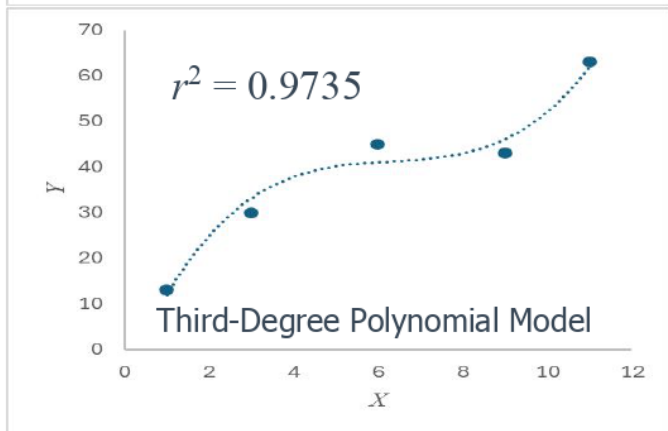
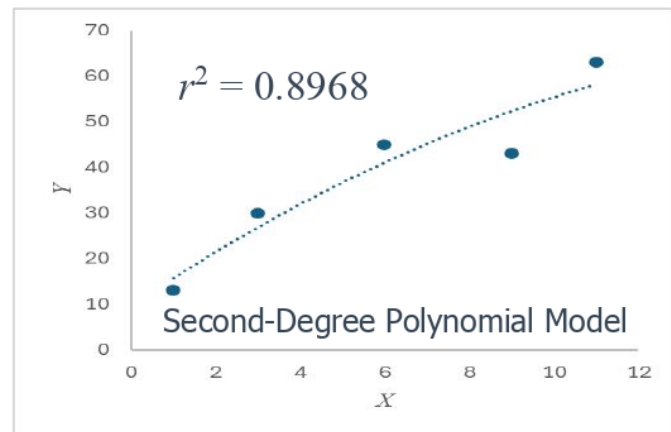
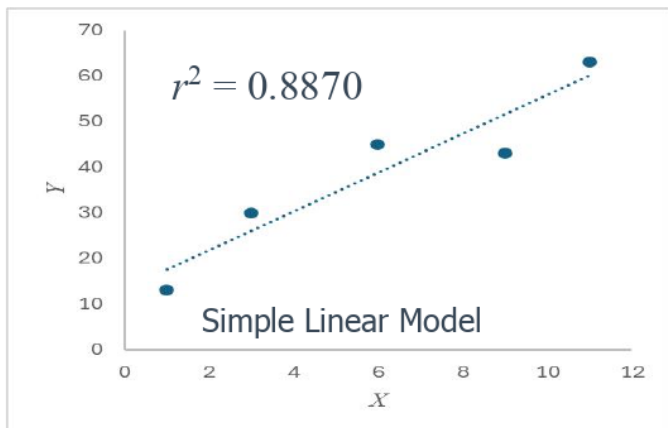
Multiple Coefficient of Determination

Note: R^2 is strongly influenced by the number of parameters included in the model.

- As the number of parameters increases ($p \rightarrow n$), the model fit generally improves, resulting in a higher R^2 (**beware of overfitting**).
- When $p = n$, the fit is perfect, but the model has no practical value.



Multiple Coefficient of Determination



Multiple Coefficient of Determination

$$SQTO = SQReg + SQE$$

Multiple coefficient of determination (r^2)

$$r^2 = \frac{SQReg}{SQTO}$$

(Avoid using this index!)

$$= 1 - \frac{SQE}{SQTO}$$

$$r_a^2 = 1 - \left(\frac{n-1}{n-p} \right) \frac{SQE}{SQTO}$$

Adjusted multiple coefficient of determination (r_a^2)

This coefficient may **decrease** if the variables added to the model do not provide meaningful contributions.

$$~~r = \sqrt{r^2}~~$$

Note: R^2 remains the standard measure of the proportion of the total variation in Y explained by the regression model.

Multicollinearity

Y	X_1	X_2	X_3	X_4
11,70	126,92	174,56	226,69	364,26
16,34	75,02	129,40	117,43	329,68
16,76	51,00	106,17	75,41	592,57
16,83	47,75	110,50	66,58	471,11
22,02	145,83	148,78	258,84	1151,11
23,43	62,91	113,04	99,85	327,56
24,75	73,34	97,81	117,23	850,26
29,96	79,87	92,83	126,21	695,32
30,31	131,55	139,24	235,10	820,23
33,51	163,68	141,01	294,77	884,83
38,12	93,25	98,44	152,29	291,09
38,42	110,57	99,38	195,38	1162,36
40,63	93,28	88,63	159,74	338,08
46,15	196,54	140,37	363,28	508,84
47,98	184,33	128,83	334,06	764,28
54,58	119,84	71,83	204,97	709,91
58,22	163,02	102,36	295,87	626,23
66,27	155,43	84,14	284,87	50,34
86,27	273,91	109,00	514,30	620,11
89,29	212,29	53,56	392,89	1186,30

$$Y_i = \beta_0 + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \beta_3 X_{3,i} + \beta_4 X_{4,i} + \xi_i$$

In general, **multicollinearity** reduces the statistical significance of the coefficient β_k associated with the independent variable k , which is correlated with one or more other independent variables. This occurs because the variance of the estimator $s^2(b_k)$ is overestimated.

Detecting multicollinearity is not always straightforward. A commonly used diagnostic is the **Variance Inflation Factor (VIF)**:

$$VIF_k = \frac{1}{1 - r_k^2}$$

where (r_k^2) is the **coefficient of determination** obtained by regressing (X_k) on all the other independent variables.

In general, if $(VIF_k > 10)$, then (X_k) exhibits severe multicollinearity.

Multicollinearity

Y	X_1	X_2	X_3	X_4
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$$Y_i = \beta_0 + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \beta_3 X_{3,i} + \beta_4 X_{4,i} + \xi_i$$

Example: Calculating the VIF for Variable (X_1)

- Fit a regression model with (X_1) as the dependent variable and all the remaining independent variables as predictors.
- Compute (R^2).
- Calculate the **VIF**.

$$X_{1,i} = a_0 + a_2 X_{2,i} + a_3 X_{3,i} + a_4 X_{4,i}$$

$$X_{1,i} = 14,04 - 0,003 X_{2,i} + 0,50 X_{3,i} + 0,0009 X_{4,i}$$

$$r_1^2 = 0,9994$$

$$VIF_1 = \frac{1}{1 - 0,9994} = 1606,4$$

- In this case, the low statistical significance of (X_1) can be explained by **multicollinearity**.
- One approach is to remove the variable with the **highest VIF** and then **repeat the analysis**.
- It is important to remove variables **one at a time** until no excessively high VIF values remain.

Eliminating Independent Variables

Y	X ₁	X ₂	X ₃	X ₄
11,70	126,92	174,56	226,69	364,26
16,34	75,02	129,40	117,43	329,68
16,76	51,00	106,17	75,41	592,57
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$$Y_i = \beta_0 + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \beta_3 X_{3,i} + \beta_4 X_{4,i} + \xi_i$$

ANOVA

	gl	SQ	MQ	F	P-value
Regression	4	9354,57	2338,64	587,45	2,78E-16
Residual	15	59,71	3,98		
Total	19	9414,28			

	Coefficients	Standard Error	Stat t	P-value
Intersection	64,4359	4,8424	13,3067	1,04E-09
X ₁	-0,2129	0,3081	-0,6908	0,5002
X ₂	-0,4741	0,0160	-29,5575	1,04E-14
X ₃	0,2659	0,1553	1,7123	0,1074
X ₄	-0,0075	0,0015	-4,8827	0,0002

Another approach is to first remove the independent variable with the **highest non-significant p-value**.

Then, **refit the model** and repeat the analysis.

Eliminating Independent Variables

Y	X_1	X_2	X_3	X_4
11,70	126,92	174,56	226,69	364,26
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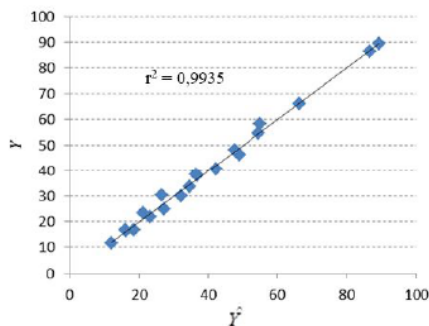
$$Y_i = \beta_0 + \beta_2 X_{2,i} + \beta_3 X_{3,i} + \beta_4 X_{4,i} + \xi_i$$

ANOVA

	gl	SQ	MQ	F	$P\text{-value}$
Regression	3	9352,67	3117,56	809,56	1,12E-17
Residual	16	61,61	3,85		
Total	19	9414,28			

	$Coefficients$	$Standard\ Error$	$Stat\ t$	$P\text{-value}$
Intersection	61,4478	2,1408	28,7030	3,44E-15
X_2	-0,4734	0,0157	-30,0637	1,66E-15
X_3	0,1587	0,0040	40,0206	1,81E-17
X_4	-0,0077	0,0015	-5,1698	9,31E-05

All coefficients are statistically significant at the 5% significance level.





QUIZ



Quiz: To evaluate the quality of a linear regression model, we should consider a combination of performance metrics. For example, the lower the RMSE and the higher the R^2 , the better the model.

TRUE

FALSE



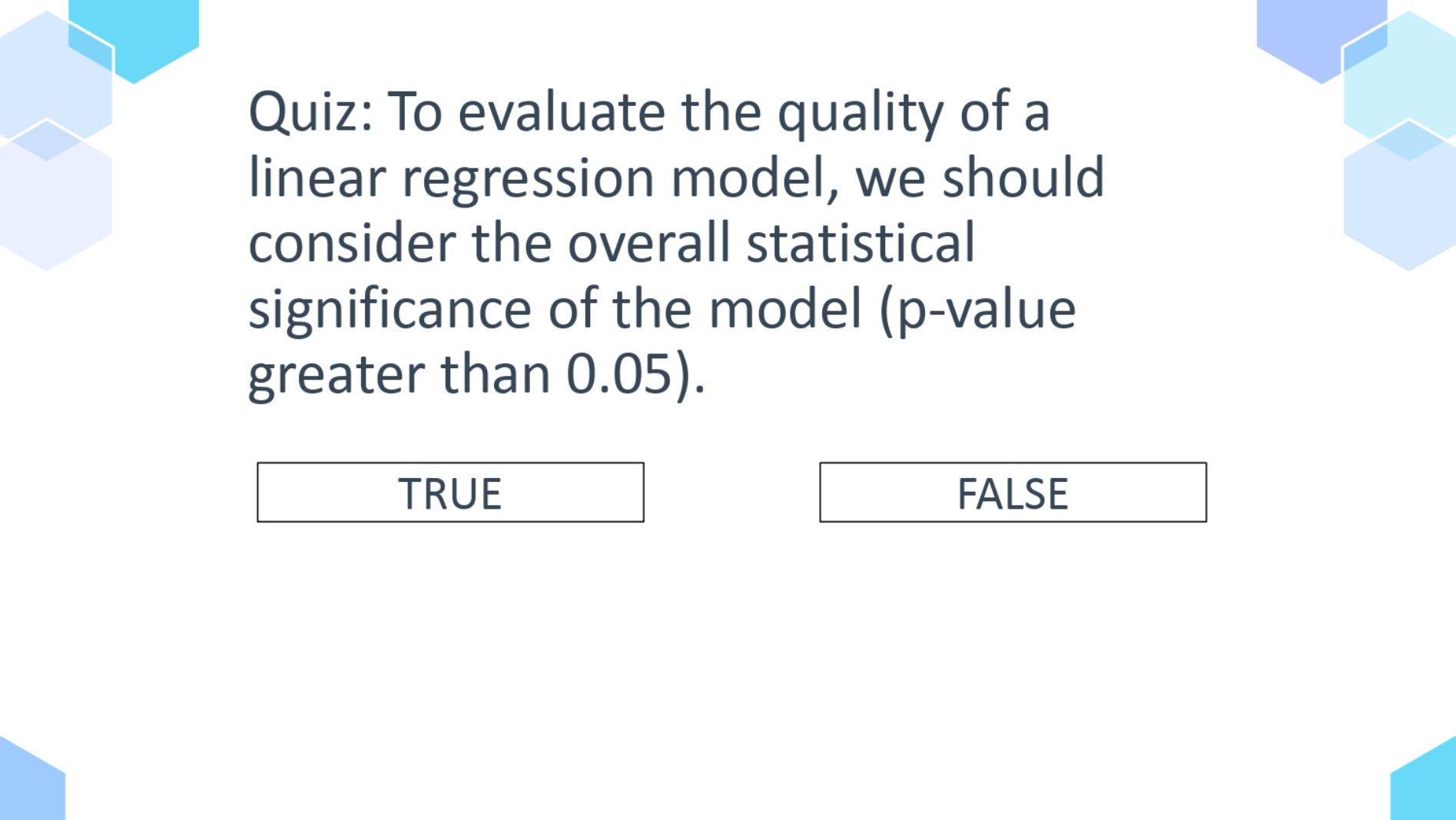


Quiz: To evaluate the quality of a linear regression model, we should consider a combination of performance metrics. For example, the lower the RMSE and the higher the R^2 , the better the model.

TRUE

FALSE





Quiz: To evaluate the quality of a linear regression model, we should consider the overall statistical significance of the model (p-value greater than 0.05).

TRUE

FALSE



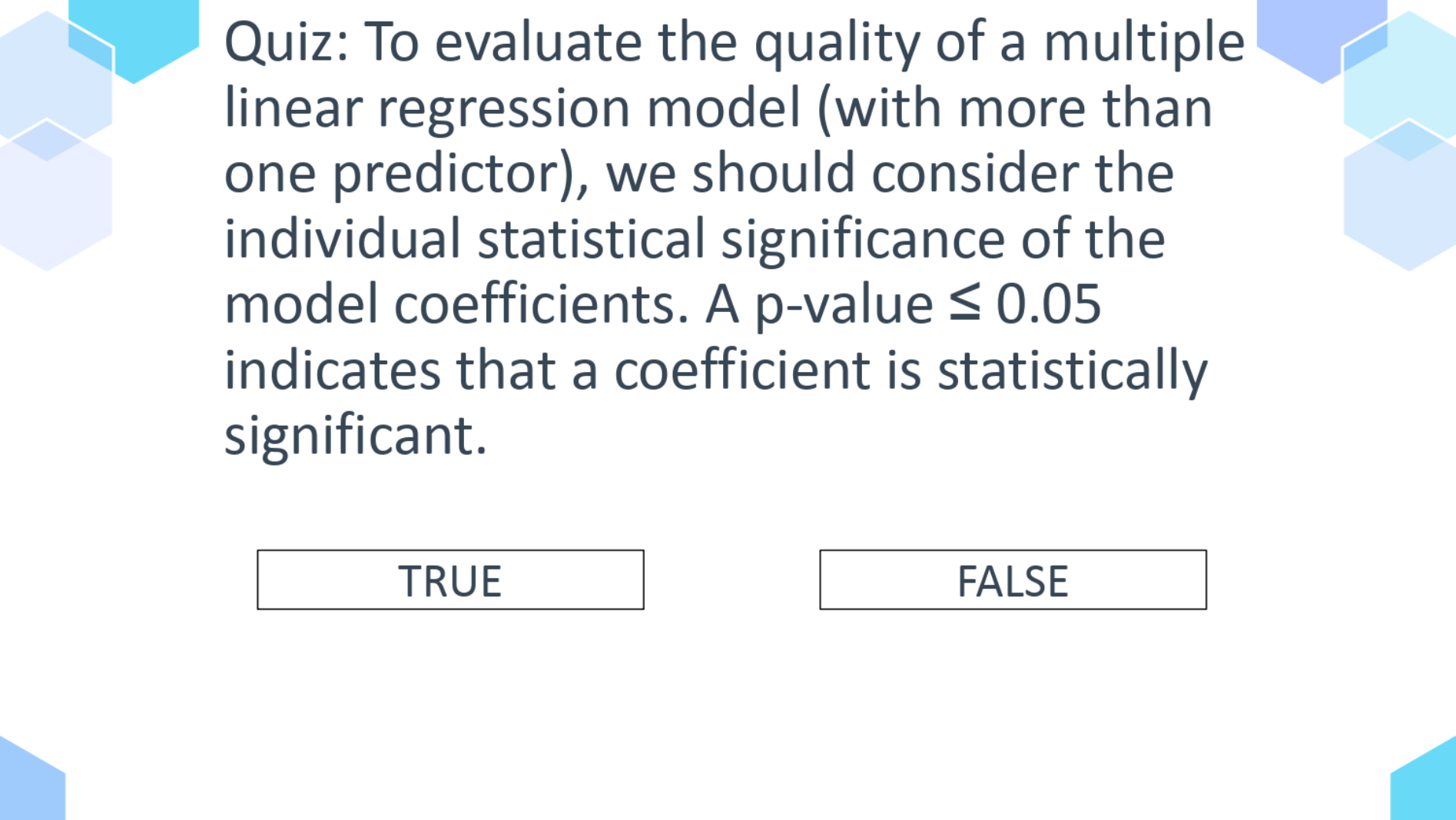
Quiz: To evaluate the quality of a linear regression model, we should consider the overall statistical significance of the model (p-value greater than 0.05).

TRUE

FALSE

The p-value must be less than or equal to 0.05.

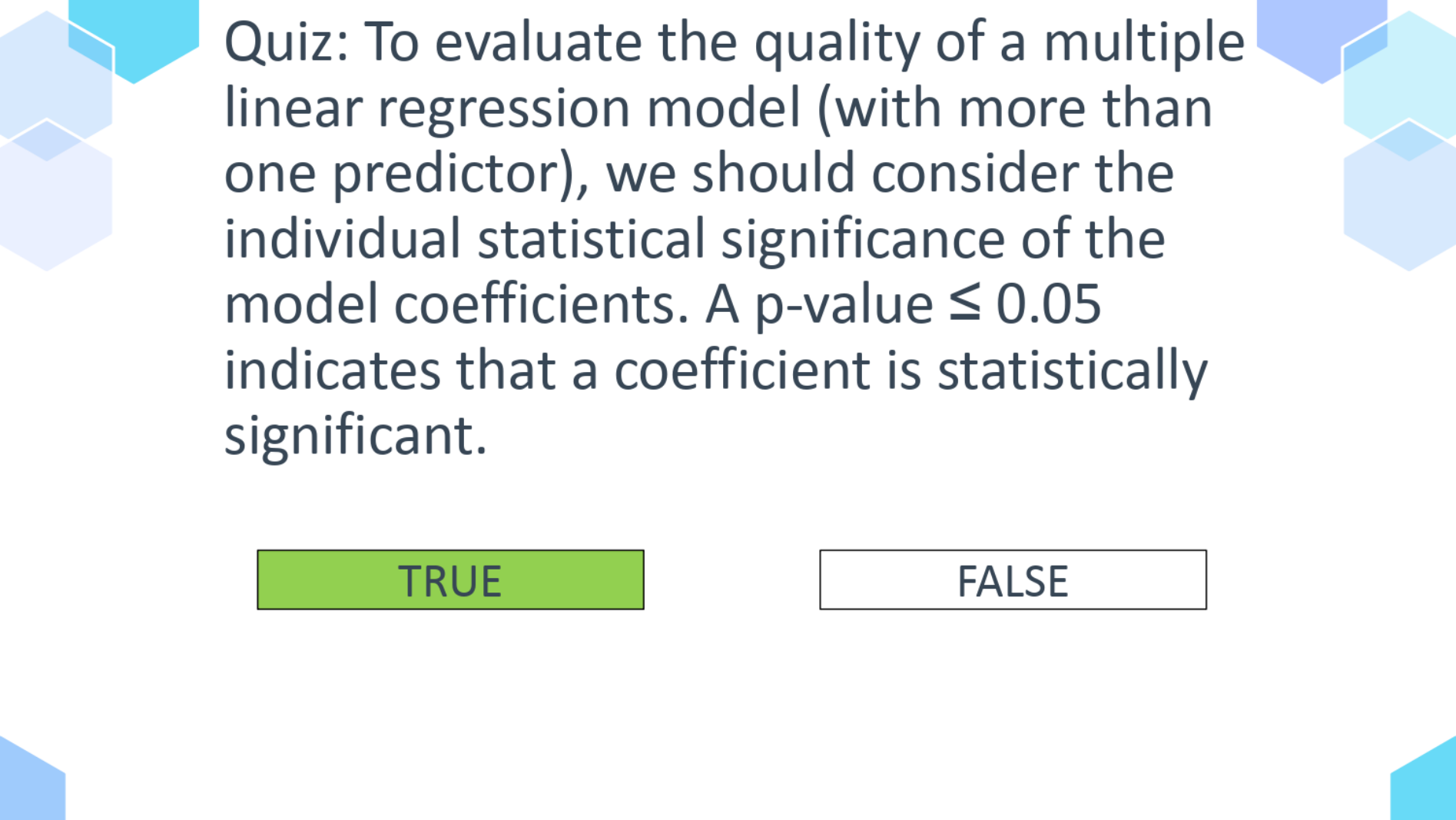




Quiz: To evaluate the quality of a multiple linear regression model (with more than one predictor), we should consider the individual statistical significance of the model coefficients. A p-value ≤ 0.05 indicates that a coefficient is statistically significant.

TRUE

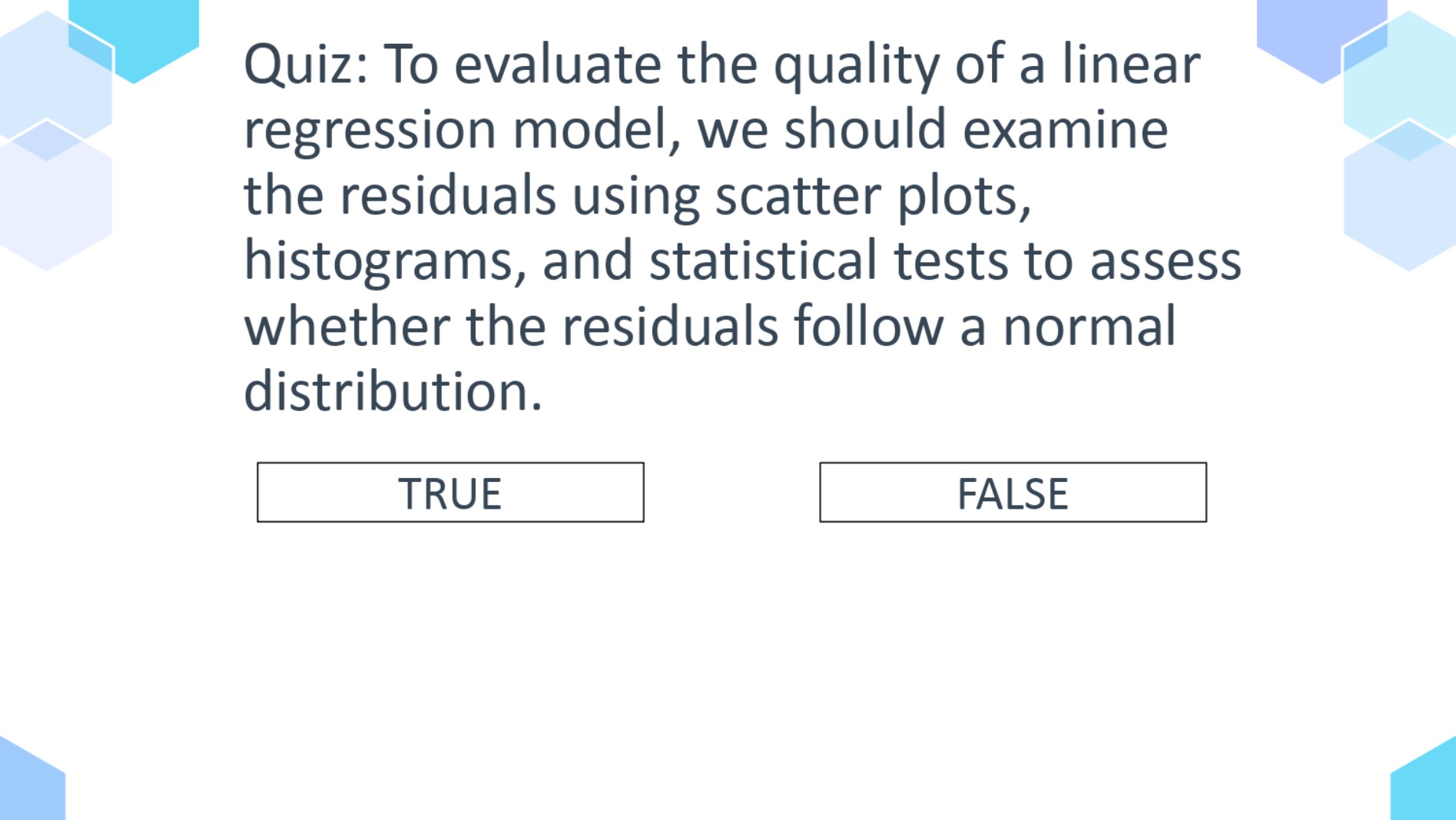
FALSE



Quiz: To evaluate the quality of a multiple linear regression model (with more than one predictor), we should consider the individual statistical significance of the model coefficients. A p-value ≤ 0.05 indicates that a coefficient is statistically significant.

TRUE

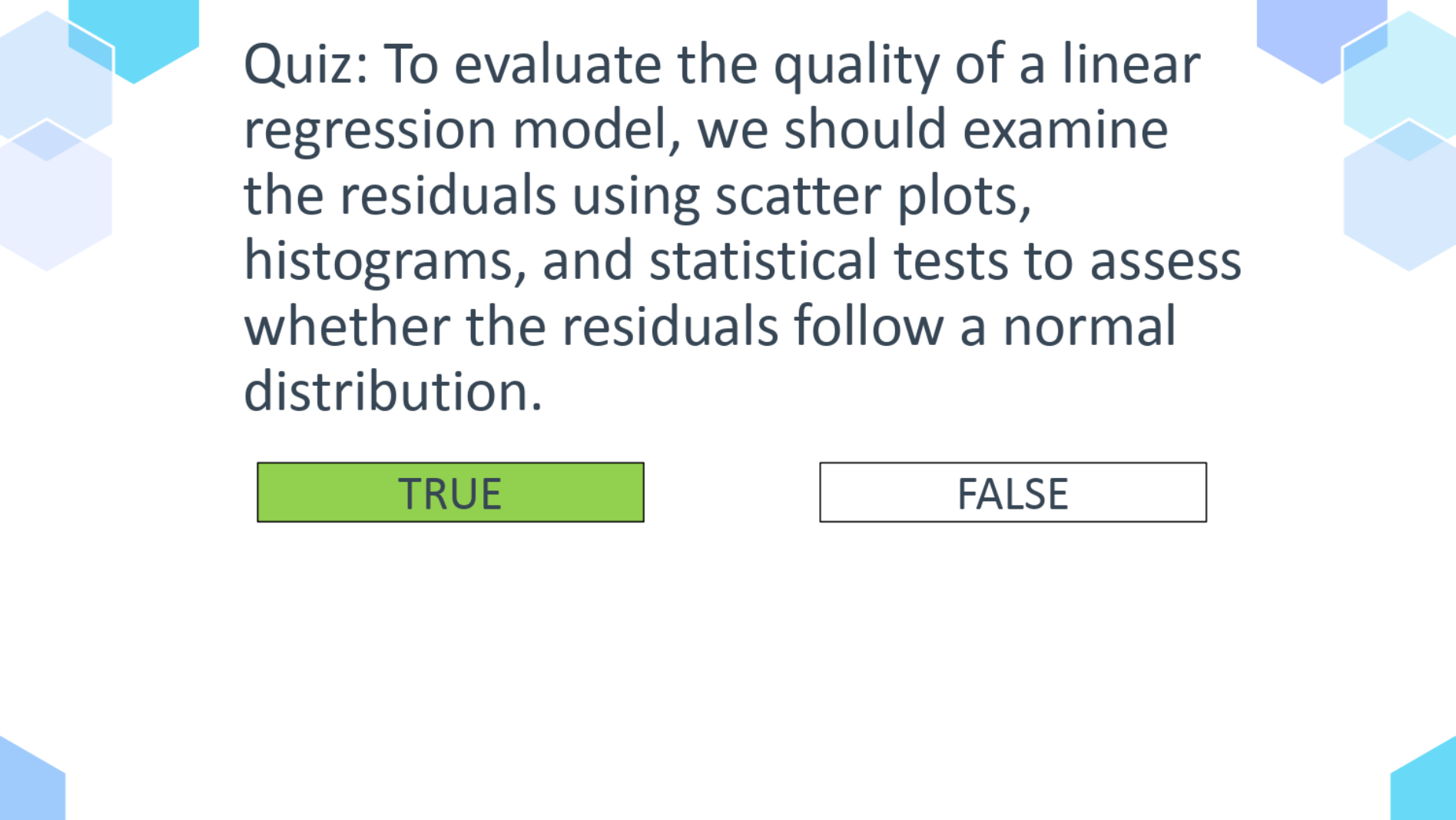
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Quiz: To evaluate the quality of a linear regression model, we should examine the residuals using scatter plots, histograms, and statistical tests to assess whether the residuals follow a normal distribution.

TRUE

FALSE



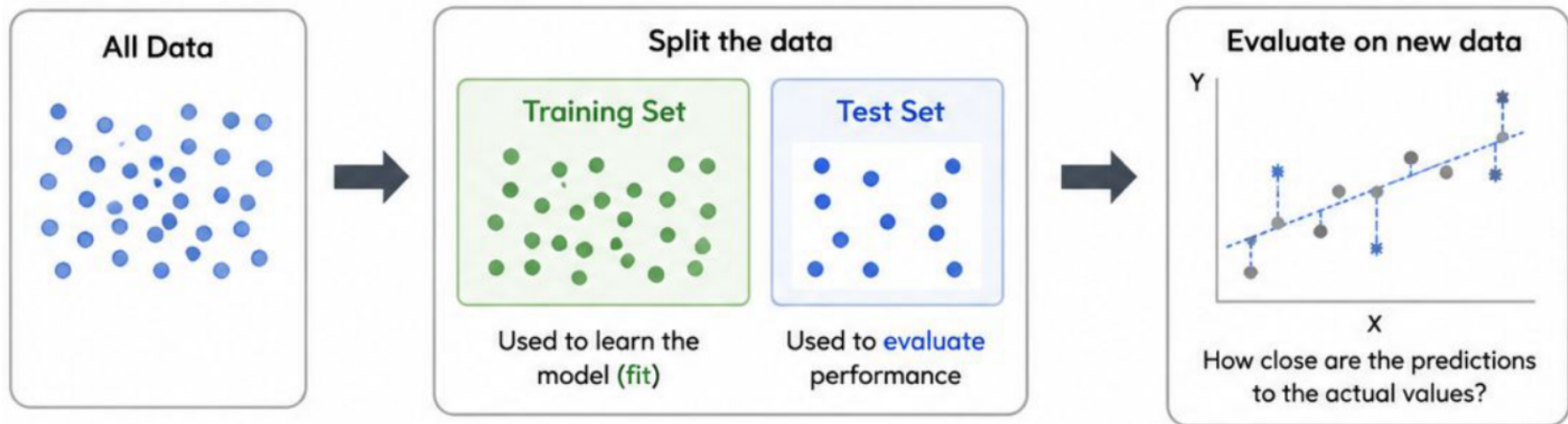
Quiz: To evaluate the quality of a linear regression model, we should examine the residuals using scatter plots, histograms, and statistical tests to assess whether the residuals follow a normal distribution.

TRUE

FALSE

How do we know if the model works on new data?

We must evaluate the model on data it *never seen* before



Goal: Build a model that **generalizes** well to **new, unseen data**.



Key idea: A model can **fit** the data it saw, but we care about how it performs on data it **has not seen**.



Next class:

Training set × Test set

How to **split** data and **evaluate** models properly.

Training Set ■■■■■■■■■■

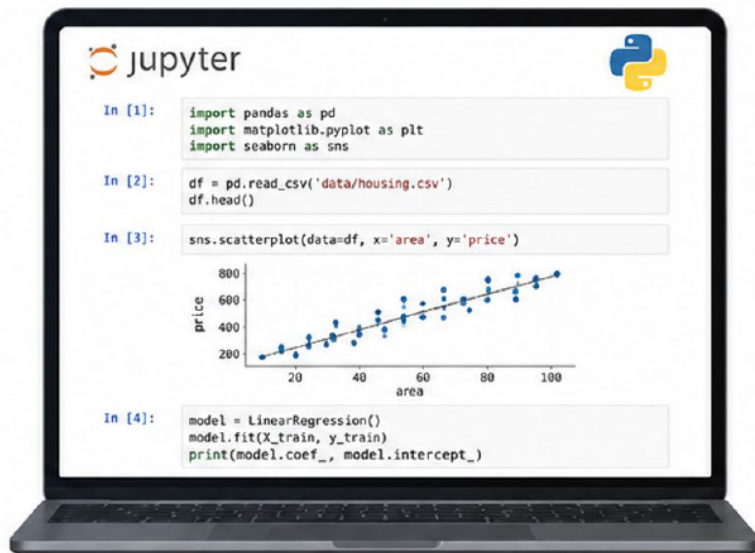
Test Set ■■■

Hands on!



What we will do

- ✓ Load and explore a real dataset
- ✓ Visualize relationships between variables
- ✓ Fit a simple and a multiple linear regression model
- ✓ Evaluate the model using metrics and residual analysis
- ✓ Interpret the results and the coefficients



Open the notebook and follow along!

We will build, evaluate, and interpret a linear regression model step by step.



Tip: Feel free to experiment, modify the code, and explore your data!

Linear Regression: Pros and Cons



PROS



Interpretability: the result of the model is an explicit equation and each coefficient has a clear meaning.



Easy to implement: fast to train, efficient, and available in virtually every statistical and machine learning software.



Statistical inference: enables hypothesis testing, confidence intervals, and p-values to understand which variables matter.



Baseline model: usually the first model tested and a strong benchmark to compare more complex algorithms.



Good performance for linear relationships: when assumptions are approximately satisfied, it often performs surprisingly well.



CONS



Assumes linearity: may not capture complex nonlinear relationships and interactions.



Sensitive to outliers: extreme values can strongly influence the estimated coefficients and predictions.



Requires several assumptions: linearity, normality of residuals, homoscedasticity, independence, and no multicollinearity.



Limited flexibility: less capable of modeling complex patterns and heterogeneity in data.

Linear Regression is often not the most accurate model, but it is frequently the most interpretable and sci informative.

Linear Regression

Édipo Cremon

